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CUET UG

Applied Mathematics

Financial Mathematics

Perpetuity, Sinking Funds, EMI, CAGR, Depreciation & Bond Valuation · Section B2 Unit VII

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Syllabus Alignment & Exam Blueprint

In the NTA CUET (UG) Applied Mathematics examination, **Financial Mathematics** (Unit VII of Section B2) forms a heavily weighted, high-scoring section comprising **6 to 8 questions** (30 to 40 marks). Every problem evaluates practical computational fluency, precise conversion of interest periods, and sharp conceptual mastery of time-value-of-money mathematics.

Sub-Topic	Official NTA Syllabus Coverage	High-Yield CUET Focus
Perpetuity & Sinking Funds	Concept of perpetuity and sinking fund; calculating ordinary and growing perpetuity; difference between sinking fund and savings account; sizing periodic sinking fund deposits.	Infinite series summation $\frac{A}{i}$; sinking fund factor $\frac{i}{(1+i)^n-1}$; earmarked legal reserve vs general savings.
EMI Computations	Equated Monthly Installments; reducing-balance method vs flat-rate method; amortization schedule; splitting installment into interest and principal components.	Converting annual rate to monthly period rate ($i = \frac{r}{12}$); loan balance recurrence; effective vs flat interest rate.
Returns, Nominal Rate & CAGR	Calculation of rate of return; nominal rate of return vs effective annual rate; Compound Annual Growth Rate (CAGR); CAGR vs Annual Growth Rate (AGR).	Capital gains plus dividend yield; geometric mean growth $\left(\frac{V_n}{V_0}\right)^{\frac{1}{n}} - 1$; volatility drag in multi-year returns.
Linear Depreciation	Linear (straight-line) depreciation method; cost, residual (scrap) value, and useful life; depreciation rate; book value schedule over asset life.	Depreciable base $(C - S)$; linear annual deduction $\frac{C-S}{n}$; book value after t years $C - tD$.
Bond Valuation	Bond terminology: Face value, coupon rate, coupon payment, maturity, Yield to Maturity (YTM); present value model of coupon bonds and zero-coupon bonds.	Bond pricing formula $V_0 = C[\text{PVIFA}] + F[\text{PVIF}]$; par vs premium vs discount conditions; inverse yield-price relation.

CUET EXAM SCORING STRATEGY — APPLIED FINANCIAL MATHEMATICS

Applied Financial Mathematics questions in CUET reward formula discipline. Over 75% of calculation mistakes arise from two pitfalls: (1) using an annual percentage rate r directly in an EMI or semi-annual bond formula without dividing by compounding frequency ($m = 12$ or $m = 2$), and (2) applying percentages to the wrong base (e.g., computing depreciation on initial cost without subtracting salvage value). Master the period-conversion rules first.

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1. Perpetuity — Valuing Infinite Cash Flows

Definition and Mathematical Formulation

A **perpetuity** is an annuity in which periodic cash flows continue indefinitely ($n \to \infty$). Since the cash flow stream has no terminal date, it has no finite future value; however, because of discounting, it possesses a finite, well-defined **Present Value (PV)**.

Let an ordinary perpetuity provide a periodic payment A at the end of each period, with discount rate i per compounding period. The present value is the sum of an infinite geometric progression:

$$\text{PV} = \frac{A}{1+i} + \frac{A}{(1+i)^2} + \frac{A}{(1+i)^3} + \ldots = \sum_{k=1}^{\infty} \frac{A}{(1+i)^k}$$

Since $\frac{1}{1+i} < 1$ for any positive interest rate $i > 0$, the sum of this infinite series converges to:

$$\text{PV} = \frac{\frac{A}{1+i}}{1 - \frac{1}{1+i}} = \frac{\frac{A}{1+i}}{\frac{i}{1+i}} = \frac{A}{i}$$

PERPETUITY VALUATION FORMULAS

1. **Ordinary Perpetuity (Payments at end of each period):**

$$\text{PV} = \frac{A}{i}$$

2. **Perpetuity Due (Payments at beginning of each period):**

$$\text{PV}_{\text{due}} = A + \frac{A}{i} = A\left(1 + \frac{1}{i}\right) = A\left(\frac{1+i}{i}\right)$$

3. **Growing Perpetuity (Cash flows grow at constant rate $g < i$ per period):**

$$\text{PV}_{\text{growing}} = \frac{A_1}{i - g}$$

where A_1 is the cash flow received at the end of Period 1.

WORKED EXAMPLE 1 — UNIVERSITY CHAIR ENDOWMENT

A philanthropist wishes to establish a perpetual scholarship providing ₹75,000 annually to meritorious students. If the endowment fund earns a steady annual return of 6%, determine:

1. The lump sum required if the first grant is awarded one year from today.
2. The lump sum required if the first grant is awarded immediately today.

Solution:

1. For an ordinary perpetuity:

$$\text{PV} = \frac{A}{i} = \frac{75000}{0.06} = 12,50,000 \text{ rupees.}$$

2. For a perpetuity due (immediate first payout):

$$\text{PV}_{\text{due}} = A + \frac{A}{i} = 75,000 + 12,50,000 = 13,25,000 \text{ rupees.}$$

Deferred Perpetuity

If an ordinary perpetuity is deferred by m periods (meaning the first payment occurs at the end of period $m+1$), its present value evaluated today ($t = 0$) is obtained by discounting the value of the perpetuity at time $t = m$:

$$\text{PV}_{\text{deferred}} = \left(\frac{A}{i}\right) \cdot \frac{1}{(1+i)^m} = \frac{A}{i(1+i)^m}$$

WORKED EXAMPLE 2 — DEFERRED RESEARCH GRANT

A research foundation commits to an endless annual grant of ₹1,20,000, with the first installment payable 4 years from today (i.e. deferred for 3 full years). At an interest rate of 8% compounded annually:

$$\text{PV} = \frac{120000}{0.08} \cdot \frac{1}{(1.08)^3} = 15,00,000 \cdot \frac{1}{1.259712} \approx 11,90,748.83 \text{ rupees.}$$

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2. Sinking Funds — Systematic Corpus Accumulation

Concept and Core Mechanics

A **sinking fund** is a special-purpose reserve fund created by setting aside periodic, equal payments (R) at compound interest (i) so that the accumulated future value equals a predetermined targeted liability (A) at maturity (n periods).

Sinking funds are mandatory financial instruments in corporate and municipal finance for:

- Redemption of long-term debentures or bonds at maturity.
- Replacement of depreciable fixed capital assets (plant, heavy machinery) at end of useful life.
- Creation of emergency contingency or environmental restoration reserves.

Sinking Fund Mathematical Derivation

Since periodic deposits R represent an ordinary annuity whose future value accumulates to target A:

$$A = R \cdot \frac{(1+i)^n - 1}{i}$$

Solving for the periodic installment R:

$$R = A \cdot \frac{i}{(1+i)^n - 1}$$

The factor $\frac{i}{(1+i)^n-1}$ is termed the **Sinking Fund Factor**.

Comparison Feature	Sinking Fund	Savings Account
Primary Objective	Earmarked contractually for a specific future financial obligation.	General-purpose liquidity, personal savings, or wealth accumulation.
Withdrawal Flexibility	Strictly locked; premature liquidation penalized or legally restricted.	Unrestricted withdrawals permitted at depositor's discretion.
Deposit Schedule	Mathematically computed fixed equal installments targeted at maturity value A.	Arbitrary deposit amounts and frequencies chosen by account holder.
Tax & Accounting Treatment	Often treated as an appropriation of profits / balance sheet reserve.	Personal asset earning taxable interest income.

WORKED EXAMPLE 3 — DEBENTURE REDEMPTION SINKING FUND

InfraCorp Ltd. issues ₹20,00,000 worth of debentures repayable in full after 5 years. A sinking fund earning 10% interest per annum compounded annually is established. Calculate the annual deposit required. (Given: $(1.10)^5 = 1.61051$).

Solution: Target corpus $A = 20,00,000$, $i = 0.10$, $n = 5$.

$$R = 20,00,000 \cdot \frac{0.10}{(1.10)^5 - 1} = 2,00,\frac{000}{1.61051 - 1} = 2,00,\frac{000}{0.61051} \approx 3,27,594.96 \text{ rupees.}$$

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3. Equated Monthly Installments (EMI)

Overview and Computational Approaches

An **Equated Monthly Installment (EMI)** is a fixed payment amount made by a borrower to a lender on a specified calendar date each month. EMIs are structured to liquidate both outstanding principal and accrued interest over a specified loan tenure.

There are two primary methods used in financial mathematics to determine EMI:

1. **Flat Rate Method:** Interest is calculated on the initial principal for the full loan term.
2. **Reducing Balance (Amortization) Method:** Interest is computed only on the outstanding principal balance at the beginning of each monthly payment cycle.

The Reducing Balance EMI Formula

Under the reducing balance method, the present value of all monthly installments E must equal the loan principal P:

$$P = E \cdot \frac{1-(1+i)^{-n}}{i} = E \cdot \frac{(1+i)^n - 1}{i(1+i)^n}$$

Solving explicitly for the monthly installment E:

$$E = P \cdot i \cdot \frac{(1+i)^n}{(1+i)^n - 1}$$

where:

- P = Loan principal borrowed.
- r = Annual interest rate; $i = \frac{r}{12 \cdot 100}$ is the **monthly interest rate**.
- t = Loan tenure in years; $n = 12t$ is the **total number of monthly installments**.

FLAT RATE VS REDUCING BALANCE EMI COMPARISON

- **Flat Rate EMI:**

$$I = P \cdot r \cdot \frac{t}{100}, \quad E_{\text{flat}} = \frac{P+I}{n} = \frac{P\left(1 + r\frac{t}{100}\right)}{12t}$$

- **Reducing Balance EMI:**

$$E_{\text{reducing}} = P \cdot i \cdot \frac{(1+i)^n}{(1+i)^n - 1}$$

Critical CUET Insight: Flat rate loans conceal the true cost of credit. Because the borrower repays principal monthly, the average principal outstanding over the tenure is roughly $\frac{P}{2}$. Consequently, an 8% flat rate corresponds to an effective reducing rate of approximately $14\% - 15\%$.

WORKED EXAMPLE 4 — VEHICLE LOAN EMI

A consumer purchases an electric car by taking a bank loan of ₹6,00,000 at 12% per annum on a reducing balance basis for a tenure of 2 years (24 months). Compute the EMI. (Given: $(1.01)^{24} = 1.26973$).

Solution: Principal $P = 6,00,000$. Monthly interest rate $i = 12\frac{\%}{12} = 1\% = 0.01$. Number of months $n = 24$.

$$E = 6,00,000 \cdot 0.01 \cdot \frac{1.26973}{1.26973 - 1} = 6000 \cdot \frac{1.26973}{0.26973} = \frac{7618.38}{0.26973} \approx 28,244.47 \text{ rupees.}$$

Loan Amortization Schedule Mechanics

In each installment t, the payment E is partitioned into two distinct components:

1. **Interest Component (I_t):** Paid on outstanding opening balance B_{t-1}:
$$I_t = B_{t-1} \cdot i$$
2. **Principal Repayment Component (P_t):** Directly reduces loan principal:
$$P_t = E - I_t$$
3. **Closing Balance (B_t):**
$$B_t = B_{t-1} - P_t$$

As the loan progresses, B_{t-1} decreases monotonically; hence I_t falls steadily, and P_t increases exponentially.

Month (t)	Opening Balance (B_{t-1})	EMI (E)	Interest ($I_t = B_{t-1}i$)	Principal ($P_t = E - I_t$)
1	₹1,00,000.00	₹17,254.80	₹1,000.00	₹16,254.80
2	₹83,745.20	₹17,254.80	₹837.45	₹16,417.35
3	₹67,327.85	₹17,254.80	₹673.28	₹16,581.52
4	₹50,746.33	₹17,254.80	₹507.46	₹16,747.34
5	₹33,998.99	₹17,254.80	₹339.99	₹16,914.81
6	₹17,084.18	₹17,254.80	₹170.84	₹17,083.96

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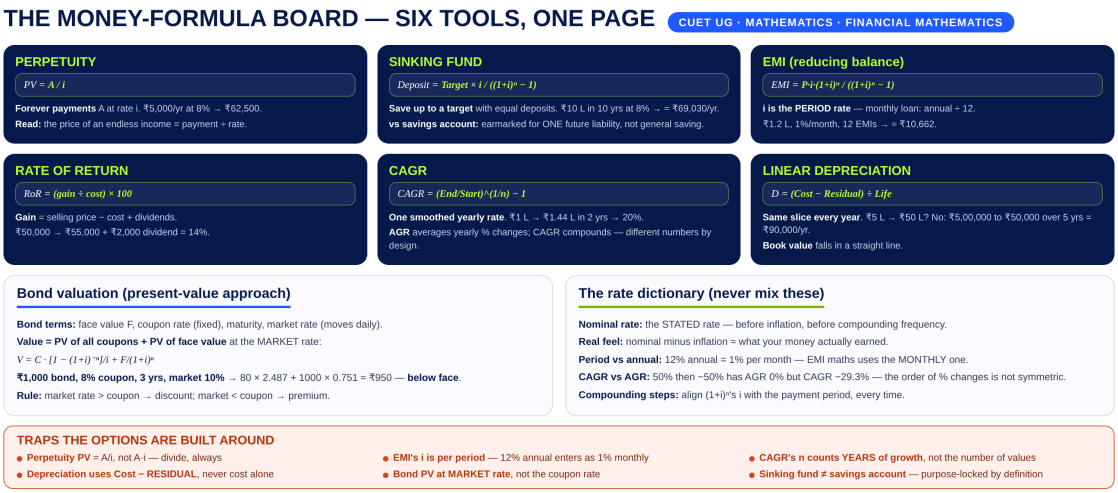


Figure 1: Figure 1: The Financial Mathematics Computational Framework — Perpetuity, Sinking Fund, EMI, Returns, CAGR & Depreciation.

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4. Returns, Nominal vs Effective Rates & CAGR

Rate of Return (RoR)

The **Rate of Return (RoR)** measures the percentage financial gain or loss generated on an investment relative to its initial cost. For any asset holding period:

$$\text{Rate of Return} = \frac{\text{Final Value} - \text{Initial Cost} + \text{Dividends / Cash Income}}{\text{Initial Cost}} \times 100$$

WORKED EXAMPLE 5 — STOCK INVESTMENT RETURN

An investor buys 100 shares of a company at ₹450 per share. Over a 1-year holding period, she receives a total dividend payout of ₹1,800. At the end of the year, she sells all shares at ₹510 per share. Determine her total rate of return.

Solution:

- Initial Investment $= 100 \times 450 = 45,000$ rupees.
- Final Value $= 100 \times 510 = 51,000$ rupees.
- Capital Gain $= 51,000 - 45,000 = 6,000$ rupees.
- Dividend Income $= 1,800$ rupees.
- Total Return $= \frac{6,000+1,800}{45,000} \times 100 = 7,\frac{800}{45},000 \times 100 = 17.33\%$.

Nominal Rate vs Effective Annual Rate (EAR)

When compounding occurs more frequently than once a year (semi-annually, quarterly, monthly), the quoted annual rate is the **nominal rate (r)**. The actual annualized rate earned due to compounding is the **Effective Annual Rate (EAR)**:

$$\text{EAR} = \left(1 + \frac{r}{m}\right)^m - 1$$

where r is the nominal annual rate expressed as a decimal, and m is compounding intervals per year.

Compounding Frequency	Periods (m)	Formula for $1 + \text{EAR}$	EAR for Nominal $r = 12\%$
Annual	1	$(1 + 0.12)^1$	12.000%
Semi-Annual	2	$(1 + 0.06)^2$	12.360%
Quarterly	4	$(1 + 0.03)^4$	12.551%
Monthly	12	$(1 + 0.01)^{12}$	12.683%
Continuous	∞	$e^{0.12}$	12.750%

Compound Annual Growth Rate (CAGR)

The **Compound Annual Growth Rate (CAGR)** provides the constant smoothed annualized geometric growth rate required for an investment to grow from its initial value to its terminal value over n years:

$$\text{Ending Value} = \text{Beginning Value} \cdot (1 + \text{CAGR})^n$$

$$\text{CAGR} = \left(\frac{\text{Ending Value}}{\text{Beginning Value}}\right)^{\frac{1}{n}} - 1$$

CAGR VS ANNUAL GROWTH RATE (AGR) — VOLATILITY DRAG

The **Annual Growth Rate (AGR)** represents the arithmetic mean of individual annual percentage returns:

$$\text{AGR} = \frac{R_1 + R_2 + \ldots + R_n}{n}$$

Because arithmetic averaging ignores compounding, **AGR will always be strictly greater than or equal to CAGR** ($\text{AGR} \geq \text{CAGR}$). The divergence between AGR and CAGR increases with annual return volatility.

- **Example:** If an investment of ₹1,00,000 gains +50% in Year 1 (to ₹1,50,000) and loses −50% in Year 2 (to ₹75,000):

$$\text{AGR} = \frac{+50\% - 50\%}{2} = 0\%$$

$$\text{CAGR} = \left(75, \frac{000}{1}, 00, 000\right)^{\frac{1}{2}} - 1 = \sqrt{0.75} - 1 = 0.8660 - 1 = -13.40\%$$

The investor suffered an actual loss of ₹25,000; CAGR accurately reflects the negative reality, while AGR provides a misleading zero-change illusion.

WORKED EXAMPLE 6 — MUTUAL FUND CAGR CALCULATION

A portfolio valued at ₹2,50,000 on 1st April 2021 grows to ₹5,18,400 on 1st April 2024 (a period of 3 years). Determine the CAGR.

Solution:

$$\text{Beginning Value} = 2, 50, 000, \quad \text{Ending Value} = 5, 18, 400, \quad n = 3.$$

$$\text{Ratio} = 5, 18, \frac{400}{2}, 50,000 = \frac{5184}{2500} = 2.0736.$$

$$\text{CAGR} = (2.0736)^{\frac{1}{3}} - 1 = (1.2754) - 1 = 27.54\% \text{ per annum.}$$

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5. Linear Method of Depreciation

Concepts and Terminology

Depreciation is the systematic allocation of the depreciable cost of a tangible fixed asset over its estimated useful economic life. In Applied Mathematics, the primary technique examined is the **Linear (Straight Line) Method**.

Core parameters:

- **Cost of Asset (C):** Purchase price plus all direct capitalized acquisition costs (freight, installation, import duties, site preparation).
- **Residual / Scrap / Salvage Value (S):** Estimated net realizable value of the asset at the conclusion of its useful life.
- **Useful Life (n):** Estimated operational period (in years) during which the asset remains economically productive.
- **Depreciable Cost / Base:**

$$\text{Depreciable Base} = C - S$$

Depreciation Formulas

LINEAR DEPRECIATION FORMULA SET

1. **Annual Depreciation Amount (D):**

$$D = \frac{C - S}{n}$$

2. **Depreciation Rate (R_D):**

$$R_D = \frac{D}{C} \times 100 = \frac{C - S}{n \cdot C} \times 100\%$$

3. **Accumulated Depreciation after t years (AccD_t):**

$$\text{AccD}_t = t \cdot D = t \cdot \left(\frac{C - S}{n}\right)$$

4. **Book Value after t years (BV_t):**

$$\text{BV}_t = C - \text{AccD}_t = C - t \cdot D$$

At the end of useful life ($t = n$): $\text{BV}_n = C - n\left(\frac{C-S}{n}\right) = S$.

WORKED EXAMPLE 7 — FACTORY MACHINERY DEPRECIATION

A manufacturing unit imports industrial packaging equipment for ₹12,00,000, paying ₹80,000 in transit insurance and ₹1,20,000 for plant commissioning. The equipment has an estimated life of 8 years, after which its scrap value is projected at ₹2,00,000. Calculate:

1. Total capitalized acquisition cost C.
2. Annual depreciation under straight line method.
3. The book value of the machinery at the end of the 5th year.

Solution:

1. $C = 12,00,000 + 80,000 + 1,20,000 = 14,00,000$ rupees.
2. Depreciable Base $= C - S = 14,00,000 - 2,00,000 = 12,00,000$ rupees.
$$D = \frac{12,00,000}{8} = 1,50,000 \text{ rupees per annum.}$$
3. Accumulated Depreciation after 5 years $= 5 \times 1,50,000 = 7,50,000$ rupees.
$$\text{BV}_5 = C - \text{AccD}_5 = 14,00,000 - 7,50,000 = 6,50,000 \text{ rupees.}$$

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6. Valuation of Bonds

Bond Terminology & Mechanics

A **bond** is a long-term debt instrument issued by corporations, municipalities, or sovereign governments to raise capital from investors. A standard coupon bond binds the issuer to pay periodic interest (coupons) and redeem the face value at maturity.

Fundamental Parameters:

- **Face Value (F):** Par value stated on the certificate (typically ₹100 or ₹1,000 in Indian bond markets), repaid at maturity.
- **Coupon Rate (c):** Contractual annual nominal interest rate printed on the bond certificate.
- **Coupon Payment (C):** Cash paid per coupon period: $C = c \cdot F$ (or $\frac{c \cdot F}{2}$ for semi-annual payments).
- **Maturity (n):** Number of periods remaining until principal redemption.
- **Yield to Maturity (YTM, r):** Required market discount rate or internal rate of return demanded by investors for bonds of comparable risk.

Intrinsic Bond Valuation Formula (Present Value Approach)

The intrinsic economic value (V_0) of a bond equals the present value of all expected future cash flows discounted at the market yield (r):

$$V_0 = \sum_{t=1}^{n} \frac{C}{(1+r)^t} + \frac{F}{(1+r)^n}$$

Applying the annuity present value factor ($\text{PVIFA}_{r,n}$) and lump sum factor ($\text{PVIF}_{r,n}$):

$$V_0 = C \cdot \left[\frac{1-(1+r)^{-n}}{r}\right] + \frac{F}{(1+r)^n}$$

WORKED EXAMPLE 8 — BOND VALUATION WITH ANNUAL COUPON

A 5-year government bond with a face value of ₹1,000 carries an 8% annual coupon. If the current market yield (YTM) for similar securities is 10%, calculate the intrinsic value of the bond. (Given: $(1.10)^{-5} = 0.62092$).

Solution:

- Coupon payment $C = 8\% \times 1000 = 80$ rupees.
- Face value $F = 1,000$ rupees.
- Discount rate $r = 10\% = 0.10$, $n = 5$.
- Present value of coupons:

$$\text{PV}_{\text{coupons}} = 80 \cdot \left[\frac{1 - 0.62092}{0.10}\right] = 80 \cdot \left[\frac{0.37908}{0.10}\right] = 80 \times 3.7908 = 303.264 \text{ rupees.}$$

- Present value of face value:

$$\text{PV}_{\text{face}} = 1000 \cdot (1.10)^{-5} = 1000 \times 0.62092 = 620.92 \text{ rupees.}$$

- Intrinsic Bond Value:

$$V_0 = 303.26 + 620.92 = 924.18 \text{ rupees.}$$

Because the market yield (10%) exceeds the coupon rate (8%), the bond trades **at a discount** ($V_0 < 1000$).

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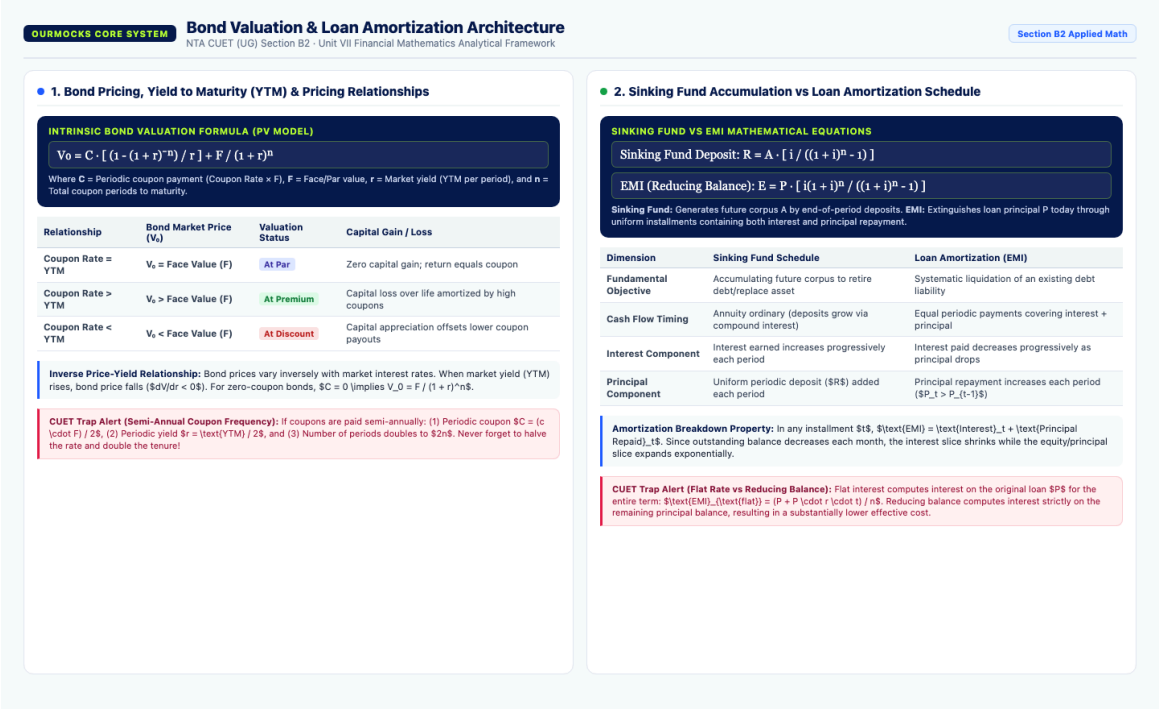


Figure 2: Figure 2: Bond Valuation Dynamics, Price-Yield Relationships & Sinking Fund vs Loan Amortization Schedules.

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Special Bond Classes

1. **Zero-Coupon Bonds (Deep Discount Bonds):** No interim coupon payments ($C = 0$). Sold at a substantial discount to par value and redeemed at par F:

$$V_0 = \frac{F}{(1+r)^n}$$

2. **Perpetual Bonds (Consols):** No maturity date ($n \to \infty$). The issuer promises an indefinite stream of coupon payments:

$$V_0 = \frac{C}{r}$$

Bond Pricing Theorems (The Three Rules of Fixed Income)

Relationship Condition	Bond Market Price (V_0)	Trading Status & Capital Return
Coupon Rate = Market YTM ($c = r$)	$V_0 =$ Face Value (F)	**Trades at Par:** Investor's entire return derives from coupon income; zero capital gain or loss.
Coupon Rate > Market YTM ($c > r$)	$V_0 >$ Face Value (F)	**Trades at Premium:** Bond price exceeds par. The investor receives above-market coupons, offset by a capital loss at maturity.
Coupon Rate < Market YTM ($c < r$)	$V_0 <$ Face Value (F)	**Trades at Discount:** Bond price is below par. Below-market coupons are compensated by capital appreciation up to par at maturity.

INVERSE RELATIONSHIP RULE

Bond prices and interest rates share an **inverse relationship**: as market interest rates (YTM) rise, bond prices fall; when market rates decline, bond prices surge ($d\frac{V_0}{d}r < 0$). Furthermore, long-maturity bonds experience significantly greater percentage price volatility in response to yield shifts than short-maturity bonds.

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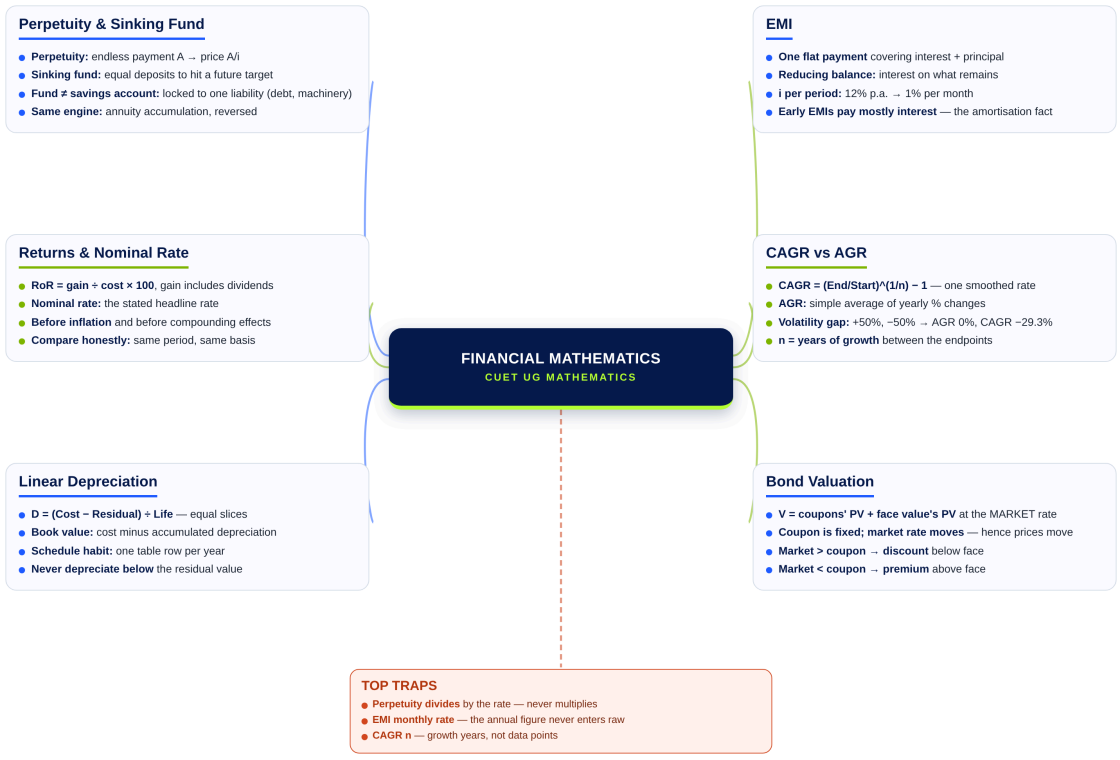


Figure 3: Figure 3: Comprehensive Applied Financial Mathematics Mindmap — Interconnecting All Six Quantitative Pillars.

HIGH-YIELD MEMORY HOOKS FOR CUET 2027

Keep these fundamental mathematical triggers primed:

- **Perpetuity:** Endless stream $\Longrightarrow$ Divide payment by period rate: PV $= \frac{A}{i}$.
- **Sinking Fund:** Future lump sum $\Longrightarrow$ Multiply target by $\frac{i}{(1+i)^n-1}$.
- **EMI:** Present loan liquidation $\Longrightarrow$ Multiply principal by $\frac{i(1+i)^n}{(1+i)^n-1}$.
- **CAGR:** Multi-year growth $\Longrightarrow$ Take n-th root of terminal-to-initial ratio minus 1.
- **SLM Depreciation:** Straight line $\Longrightarrow$ Depreciate net base $(C-S)$ equally over n years.
- **Bonds:** $V_0 =$ PV of Coupons + PV of Par. If yield exceeds coupon, bond trades at discount.

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MASTER FORMULA & DECISION CHEATSHEET — FINANCIAL MATHEMATICS

1. Perpetuity Equations:

$$\text{Ordinary Perpetuity:}\quad \text{PV} = \frac{A}{i}, \qquad \text{Perpetuity Due:}\quad \text{PV}_{\text{due}} = A + \frac{A}{i} = A\left(\frac{1+i}{i}\right)$$

$$\text{Growing Perpetuity:}\quad \text{PV} = \frac{A_1}{i-g}\quad (i>g), \qquad \text{Deferred Perpetuity:}\quad \text{PV} = \frac{A}{i(1+i)^m}$$

2. Sinking Fund Annuity:

$$R = A \cdot \frac{i}{(1+i)^n - 1} \qquad (R = \text{Periodic Deposit}, A = \text{Target Future Corpus})$$

3. Loan Amortization & EMI:

$$E_{\text{reducing}} = P \cdot i \cdot \frac{(1+i)^n}{(1+i)^n - 1}, \qquad E_{\text{flat}} = \frac{P\left(1 + r\frac{t}{100}\right)}{12t}$$

$$\text{Interest in Month } t: I_t = B_{t-1} \cdot i, \qquad \text{Principal Repaid in Month } t: P_t = E - I_t$$

4. Return & Growth Metrics:

$$\text{RoR} = \frac{\text{Ending Value} - \text{Cost} + \text{Dividends}}{\text{Cost}} \times 100\%$$

$$\text{EAR} = \left(1 + \frac{r}{m}\right)^m - 1, \qquad \text{CAGR} = \left(\frac{\text{Ending Value}}{\text{Beginning Value}}\right)^{\frac{1}{n}} - 1$$

5. Straight Line Depreciation:

$$D = \frac{C-S}{n}, \qquad R_D = \frac{D}{C} \times 100\%, \qquad \text{BV}_t = C - t \cdot D$$

6. Bond Valuation Model:

$$V_0 = C \cdot \left[\frac{1-(1+r)^{-n}}{r}\right] + \frac{F}{(1+r)^n}, \qquad V_0^{\text{zero}} = \frac{F}{(1+r)^n}, \qquad V_0^{\text{perpetual}} = \frac{C}{r}$$

THE SEVEN FATAL CUET TRAPS IN FINANCIAL MATHEMATICS

1. **The Compounding Rate Frequency Blindness:** Quoting an annual rate (e.g. 12%) in an EMI calculation without dividing by 12 ($i = 0.01$) or in a semi-annual bond without dividing by 2 ($r = 0.06$).

2. **Salvage Value Omission in Depreciation:** Computing annual depreciation as $\frac{C}{n}$ rather than $\frac{C-S}{n}$. Remember: scrap value is never depreciated!
3. **Depreciation Rate Denominator Confusion:** Calculating depreciation rate as $\frac{D}{C-S}$ instead of $\frac{D}{C}$. The official statutory rate is always expressed as a percentage of initial acquisition cost C.
4. **Flat Rate vs Reducing Balance Deception:** Assuming a flat 10% loan is cheaper than a 14% reducing loan. Because principal is paid down monthly, a 10% flat rate has an effective reducing APR near 18%!
5. **AGR vs CAGR Substitution Error:** Arithmetic averaging of volatile percentage returns (e.g., +20% and −20%) gives 0%, whereas the compounding CAGR is $\sqrt{1.2 \times 0.8} - 1 = \sqrt{0.96} - 1 = -2.02\%$.
6. **Bond Par vs Coupon vs Yield Disconnect:** Forgetting that coupon rate is fixed on face value F, while market price fluctuates based on market yield. When $r > c$, $V_0 < F$ (Discount); when $r < c$, $V_0 > F$ (Premium).
7. **Sinking Fund vs Savings Account Purpose:** Confusing general cash reserves with sinking funds. A sinking fund is contractually designated to retire a specific future debt and cannot be used for operational expenses.

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OurMocks

Practice Questions — Part 1 (Q1 to Q5)

Q1. A philanthropic trust intends to institute an annual prize of ₹18,000 in perpetuity to the topper of an engineering college. If the corpus fund is invested in government securities yielding an annual return of 7.5%, the minimum lump sum donation required today is:

(A) ₹2,25,000 **(B)** ₹2,40,000
(C) ₹1,35,000 **(D)** ₹2,50,000

Sol. Q1. For an ordinary perpetuity, the present value of endless periodic payouts A discounted at annual rate i is:

$$\text{PV} = \frac{A}{i}$$

Here, $A = 18,000$ and $i = 7.5\% = 0.075 = \frac{75}{1000} = \frac{3}{40}$.

$$\text{PV} = \frac{18000}{\frac{3}{40}} = 18000 \times \frac{40}{3} = 6000 \times 40 = 2,40,000 \text{ rupees.}$$

Hence, a lump sum endowment of ₹2,40,000 invested at 7.5% generates exactly ₹18,000 each year forever while preserving the corpus intact. **Answer: (B)**

Q2. A company needs to replace a transport fleet in 4 years at an estimated cost of ₹15,00,000. It establishes a sinking fund earning 8% interest per annum compounded annually. The equal annual deposit required at the end of each year is: (Given: $(1.08)^4 = 1.3605$)

(A) ₹3,75,000.00 **(B)** ₹3,32,871.01
(C) ₹4,08,150.00 **(D)** ₹3,00,000.00

Sol. Q2. Target future corpus $A = 15,00,000$, interest rate $i = 0.08$, tenure $n = 4$ years. The periodic sinking fund deposit R is given by:

$$R = A \cdot \frac{i}{(1+i)^n - 1}$$

Substituting the given value $(1.08)^4 = 1.3605$:

$$R = 15,00,000 \cdot \frac{0.08}{1.3605 - 1} = 1,20,\frac{000}{0.3605} \approx 3,32,871.01 \text{ rupees.}$$

(Note: Option A incorrectly divides ₹15,00,000 simply by 4 without taking interest accumulation into account). **Answer: (B)**

Q3. A housing finance company offers a home improvement loan of ₹3,00,000 repayable in 36 equated monthly installments at 12% per annum on a reducing balance basis. The monthly interest rate (i) and the number of periods (n) to be utilized in the EMI formula are:

(A) $i = 0.12, \quad n = 3$ **(B)** $i = 0.01, \quad n = 36$
(C) $i = 0.06, \quad n = 6$ **(D)** $i = 0.001, \quad n = 36$

Sol. Q3. In monthly installment loan calculations:

- The annual nominal interest rate $r = 12\%$ must be converted to a monthly periodic rate:

$$i = \frac{r}{12 \cdot 100} = \frac{12}{1200} = 0.01 \quad \text{(or 1\% per month).}$$

- The total number of periods $n = \text{Tenure in years} \times 12 = 3 \times 12 = 36$ months.

Matches Option B precisely. Using $i = 0.12$ directly is a classic trap. **Answer: (B)**

Q4. The primary legal and structural difference between a sinking fund and an ordinary savings bank account is that a sinking fund:

(A) Pays no interest on accumulated balances **(B)** Is legally earmarked for the systematic liquidation of a specific predetermined future obligation
(C) Allows unlimited daily debit withdrawals via automated teller machines **(D)** Is exclusively available to non-profit charitable societies

Sol. Q4. A sinking fund is a specialized financial device whose sole objective is accumulating capital to retire a specific designated liability (e.g., bond redemption or plant replacement) at a fixed maturity date. Unlike general savings accounts, sinking fund withdrawals are strictly restricted to that designated purpose. **Answer: (B)**

Q5. Under the reducing balance method of loan amortization, which of the following statements correctly characterizes the behavior of the interest and principal components across the loan tenure?

(A) Both interest and principal components remain strictly constant in every installment. **(B)** The interest component increases progressively while the principal component decreases.
(C) The interest component decreases progressively while the principal component increases. **(D)** The total installment amount (E) fluctuates depending on monthly bank base rate revisions.

Sol. Q5. In an amortization schedule with fixed EMI E:

$$E = I_t + P_t$$

Because each payment partially pays down the outstanding principal balance, the opening balance B_{t-1} drops each month. Consequently, the interest component $I_t = B_{t-1} \cdot i$ decreases monotonically. To keep the total EMI E constant, the principal repayment component $P_t = E - I_t$ must progressively increase. **Answer: (C)**

Practice Questions — Part 2 (Q6 to Q10)

Q6. An individual purchased 200 shares of a pharmaceutical firm at ₹350 per share. Over the course of 12 months, she received a dividend of ₹15 per share. At the end of the year, she liquidated her holding at ₹406 per share. Her total rate of return is:

(A) 16.0% **(B)** 20.3%
(C) 11.7% **(D)** 24.5%

Sol. Q6. Total Initial Cost $= 200 \times 350 = 70,000$ rupees. Sale Proceeds $= 200 \times 406 = 81,200$ rupees. Capital Gain $= 81,200 - 70,000 = 11,200$ rupees. Total Dividend Income $= 200 \times 15 = 3,000$ rupees. Total Gain $= 11,200 + 3,000 = 14,200$ rupees.

$$\text{Rate of Return} = \left(\frac{\text{Total Gain}}{\text{Total Initial Cost}}\right) \times 100 = \left(14, \frac{200}{70}, 000\right) \times 100 = 20.2857\% \approx 20.3\%.$$

Answer: (B)

Q7. A venture capital investment in an enterprise grew from an initial valuation of ₹4,00,000 to ₹13,50,000 over a period of 3 years. The Compound Annual Growth Rate (CAGR) of this venture is: (Given: $(3.375)^{\frac{1}{3}} = 1.50$)

(A) 79.2% **(B)** 50.0%
(C) 42.5% **(D)** 33.3%

Sol. Q7. Initial value $V_0 = 4,00,000$, final value $V_3 = 13,50,000$, tenure $n = 3$ years.

$$\text{CAGR} = \left(\frac{V_3}{V_0}\right)^{\frac{1}{n}} - 1 = \left(13, 50, \frac{000}{4}, 00, 000\right)^{\frac{1}{3}} - 1 = \left(\frac{135}{40}\right)^{\frac{1}{3}} - 1 = (3.375)^{\frac{1}{3}} - 1$$

Using the given cube root:

$$\text{CAGR} = 1.50 - 1 = 0.50 = 50.0\%.$$

Check: $4,00,000 \times (1.5)^3 = 4,00,000 \times 3.375 = 13,50,000$ ✓. **Answer: (B)**

Q8. An equity portfolio gained +40% during its first year of operation and lost −30% during its second year. The Annual Growth Rate (AGR) and Compound Annual Growth Rate (CAGR) are respectively:

(A) AGR = 5.0%, CAGR ≈ −1.0% **(B)** AGR = 10.0%, CAGR = 5.0%
(C) AGR = 5.0%, CAGR = 5.0% **(D)** AGR = −1.0%, CAGR = 5.0%

Sol. Q8.

1. Annual Growth Rate (Arithmetic Mean):

$$\text{AGR} = \frac{R_1 + R_2}{2} = \frac{+40\% - 30\%}{2} = 10\frac{\%}{2} = 5.0\%.$$

2. Compound Annual Growth Rate: Starting with $V_0 = 100$:
   - After Year 1: $100 \times 1.40 = 140$.
   - After Year 2: $140 \times (1 - 0.30) = 140 \times 0.70 = 98$.

$$\text{CAGR} = \left(\frac{98}{100}\right)^{\frac{1}{2}} - 1 = \sqrt{0.98} - 1 \approx 0.98995 - 1 = -0.01005 \approx -1.0\%.$$

The portfolio suffered an actual net loss of 2%, which CAGR faithfully reports as a −1.0% annual drop, while AGR falsely shows a +5.0% profit. **Answer: (A)**

Q9. A high-precision CNC machine was procured for ₹10,00,000. Its estimated useful economic life is 6 years, with an anticipated scrap value of ₹1,00,000. Under the linear (straight line) method, the annual depreciation and the book value at the end of the 4th year are:

(A) Annual Depreciation = ₹1,50,000; Book Value = ₹4,00,000 **(B)** Annual Depreciation = ₹1,66,667; Book Value = ₹3,33,333
(C) Annual Depreciation = ₹1,50,000; Book Value = ₹4,50,000 **(D)** Annual Depreciation = ₹1,80,000; Book Value = ₹2,80,000

Sol. Q9. Acquisition cost $C = 10,00,000$, Scrap value $S = 1,00,000$, Useful life $n = 6$ years.

1. Depreciable Base $= C - S = 10,00,000 - 1,00,000 = 9,00,000$ rupees.
2. Annual Depreciation:

$$D = \frac{C - S}{n} = 9,00, \frac{000}{6} = 1,50,000 \text{ rupees per year.}$$

3. Accumulated Depreciation after 4 years:

$$\text{AccD}_4 = 4 \times D = 4 \times 1,50,000 = 6,00,000 \text{ rupees.}$$

4. Book Value at end of Year 4:

$$\text{BV}_4 = C - \text{AccD}_4 = 10,00,000 - 6,00,000 = 4,00,000 \text{ rupees.}$$

Answer: (A)

Q10. A 10-year corporate debenture with a face value of ₹1,000 offers a fixed coupon rate of 9% payable annually. If the required market yield (YTM) for similar instruments is currently 7.5%, the bond will trade in the secondary debt market:

(A) At par (equal to ₹1,000) **(B)** At a premium (strictly greater than ₹1,000)
(C) At a discount (strictly less than ₹1,000) **(D)** At zero value due to interest rate risk

Sol. Q10. When the contractual coupon rate of a bond ($c = 9\%$) is strictly greater than the current market required yield ($r = 7.5\%$), the bond pays higher annual cash flow than freshly issued securities of identical risk. Investors willingly bid up its market price above face value ($V_0 > F$). Therefore, the bond trades **at a premium**. **Answer: (B)**

Practice Questions — Part 3 (Q11 to Q15)

Q11. A zero-coupon bond has a face value of ₹10,000 and matures after 3 years. If the prevailing market interest rate is 8% compounded annually, the fair present market value of the bond is: (Given: $(1.08)^{-3} = 0.79383$)

(A) ₹7,938.30 (B) ₹8,000.00
(C) ₹9,200.00 (D) ₹6,830.13

Sol. Q11. For a zero-coupon bond, there are no interim coupon payments ($C = 0$). The valuation formula reduces to the present value of the face value discounted from maturity:

$$V_0 = \frac{F}{(1+r)^n} = F \cdot (1+r)^{-n}$$

Given $F = 10,000$, $r = 0.08$, $n = 3$, and $(1.08)^{-3} = 0.79383$:

$$V_0 = 10,000 \times 0.79383 = 7,938.30 \text{ rupees.}$$

Answer: (A)

Q12. A perpetual consol bond pays an annual coupon of ₹600. If the market interest rate rises from 5% to 6%, the percentage change in the market price of the consol is:

(A) Increases by 20.0% (B) Decreases by 16.67%
(C) Decreases by 20.0% (D) Remains unchanged

Sol. Q12. The price of a perpetual bond is given by:

$$V_0 = \frac{C}{r}$$

- When $r_1 = 5\% = 0.05$:

$$V_1 = \frac{600}{0.05} = 12,000 \text{ rupees.}$$

- When $r_2 = 6\% = 0.06$:

$$V_2 = \frac{600}{0.06} = 10,000 \text{ rupees.}$$

- Change in price $= V_2 - V_1 = 10,000 - 12,000 = -2,000$ rupees.
- Percentage Change $= \left(-2, \frac{000}{12}, 000\right) \times 100 = -\frac{1}{6} \times 100 = -16.67\%$.

Thus, the market price **decreases by 16.67%**, illustrating the inverse relationship between interest rates and bond valuations. **Answer: (B)**

Q13. A machine costing ₹5,00,000 is depreciated at 10% per annum using the straight-line method. If the scrap value at the end of its useful life is ₹50,000, the estimated useful life of the machine is:

(A) 9 years (B) 10 years
(C) 8 years (D) 12 years

Sol. Q13. In financial accounting and linear depreciation, the annual rate of depreciation $R_D = 10\%$ applies to the original asset cost $C = 5,00,000$:

$$D = C \times \frac{R_D}{100} = 5,00,000 \times 0.10 = 50,000 \text{ rupees per annum.}$$

The total depreciable amount over the machine's life is:

$$C - S = 5,00,000 - 50,000 = 4,50,000 \text{ rupees.}$$

Since $D = \frac{C-S}{n}$:

$$n = \frac{C-S}{D} = 4,50, \frac{000}{50}, 000 = 9 \text{ years.}$$

Answer: (A)

Q14. An investor considers two savings instruments: Scheme X offers 12% per annum compounded quarterly, while Scheme Y offers 12% per annum compounded monthly. Which scheme provides the higher Effective Annual Rate (EAR)?

(A) Scheme X, because quarterly compounding provides larger individual interest increments.
(B) Scheme Y, because more frequent compounding yields higher compound interest accumulation.
(C) Both schemes provide identical effective annual rates since nominal rates are both 12%.
(D) The EAR cannot be evaluated without knowing the absolute investment principal.

Sol. Q14. The Effective Annual Rate formula is:

$$\text{EAR} = \left(1 + \frac{r}{m}\right)^m - 1$$

As compounding frequency m increases, the effective rate strictly increases:

- For Scheme X ($m = 4$):

$$\text{EAR}_X = \left(1 + \frac{0.12}{4}\right)^4 - 1 = (1.03)^4 - 1 = 1.12551 - 1 = 12.551\%.$$

- For Scheme Y ($m = 12$):

$$\text{EAR}_Y = \left(1 + \frac{0.12}{12}\right)^{12} - 1 = (1.01)^{12} - 1 = 1.12683 - 1 = 12.683\%.$$

Scheme Y yields $12.683\% > 12.551\%$. Hence, Scheme Y provides the higher EAR. **Answer: (B)**

Q15. A borrower repays a personal loan of ₹1,00,000 taken at 12% p.a. (reducing balance) in 12 monthly installments with an EMI of ₹8,884.88. The interest component paid in the very first monthly installment is:

(A) ₹1,200.00 (B) ₹1,000.00
(C) ₹7,884.88 (D) ₹888.49

Sol. Q15. In the first monthly installment:

- Opening loan balance $B_0 = 1,00,000$ rupees.
- Monthly interest rate $i = 12\frac{\%}{12} = 1\% = 0.01$.
- Interest component I_1 in Month 1 is:

$$I_1 = B_0 \cdot i = 1,00,000 \times 0.01 = 1,000 \text{ rupees.}$$

(The remaining $P_1 = 8,884.88 - 1,000 = 7,884.88$ goes toward reducing loan principal). Option A incorrectly applies the full annual rate (12%) to the single month. **Answer: (B)**